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Mechanistic and machine learning insights into borrowing hydrogen reactions catalyzed by transition metal complexes with N-heterocyclic ligands

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Pyrazole-based transition metal complexes have attracted increasing attention in borrowing hydrogen (BH) reactions, particularly when employing the metal–ligand cooperation (MLC) strategy to achieve high activity and selectivity. However, a systematic understanding of their mechanistic selectivity and the factors governing catalytic performance remains lacking. Herein, the BH reaction of alcohols catalyzed by such complexes was investigated using density functional theory (DFT) calculations in combination with machine learning (ML) methods. Three possible pathways—N2-site-assisted, O-site-assisted, and N1-site-assisted—were proposed, among which the N2-site-assisted route was identified as the most favorable. Both the dehydrogenation and hydrogenation steps proceed *via* an outer-sphere concerted transfer mechanism. Distortion/interaction analysis revealed that the ligand-assisted distortion energy plays a decisive role in determining the activation barrier. Furthermore, an ML model with high predictive accuracy (R^2 = 0.9570) was established to correlate catalytic performance with electronic and steric descriptors. Feature importance analysis identified the HOMO energy level, dipole moment, and molecular volume as key factors, reflecting the roles of electron-donating ability, transition-state polarization, and steric effects, respectively. This study not only deepens the mechanistic understanding of MLC-enabled BH reactions catalyzed by pyrazole-based transition metal complexes but also provides a predictive framework for the rational design of efficient and tunable catalysts.

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Introduction

A borrowing hydrogen (BH) reaction is a tandem transformation catalyzed by transition metals, involving dehydrogenation of alcohols, condensation, and subsequent reduction.[1–3] It has been widely applied in the green synthesis of C–C and C–N bonds.[4–6] In this process, alcohols serve as alkylating agents and are first dehydrogenated at the metal center to generate carbonyl intermediates, which then undergo base-promoted condensation with nucleophilic substrates to form α,β-unsaturated intermediates. These intermediates are finally reduced catalytically to yield the desired products.[7,8] The borrowing hydrogen strategy offers high atom economy, operational simplicity, and water as the sole byproduct, making it an efficient and sustainable approach for synthesizing value-added organic molecules.[9–11] It is particularly well-suited for α-alkylation reactions involving alcohols and ketones or nitriles.

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In recent years, metal–ligand cooperativity (MLC) has attracted widespread attention as an effective strategy to modulate catalytic activity and selectivity in homogeneous catalysis.[12,13] This mechanism involves the cooperative participation of proton-responsive ligand groups (such as NH and OH) and the metal center in hydrogen transfer processes, effectively lowering reaction barriers and enhancing catalytic efficiency. Pyrazole-based ligands, owing to their multidentate coordination ability and excellent electronic tunability, have become important platforms for studying MLC and are widely applied in various catalytic systems including dehydrogenation, hydrogenation, and borrowing hydrogen reactions.[14,15]

In 2010, Ikariya *et al.* reported a Cp*Ir (Cp* = pentamethylcyclopentadienyl anion) complex containing a proton-responsive pyridyl ring (Fig. 1a, **Cat. 1**).[16] Under basic conditions, the catalyst undergoes reversible tautomerization between the pyrazole and its anionic form, facilitated by a hydrogen-bonding network that promotes proton transfer and strengthens metal–ligand cooperativity. This catalyst demonstrated high activity in hydroamination reactions, indicating the critical role of the β-NH site in substrate activation. In 2017, Himeda *et al.* synthesized a series of Cp*Ir complexes bearing pyridyl-pyrazole

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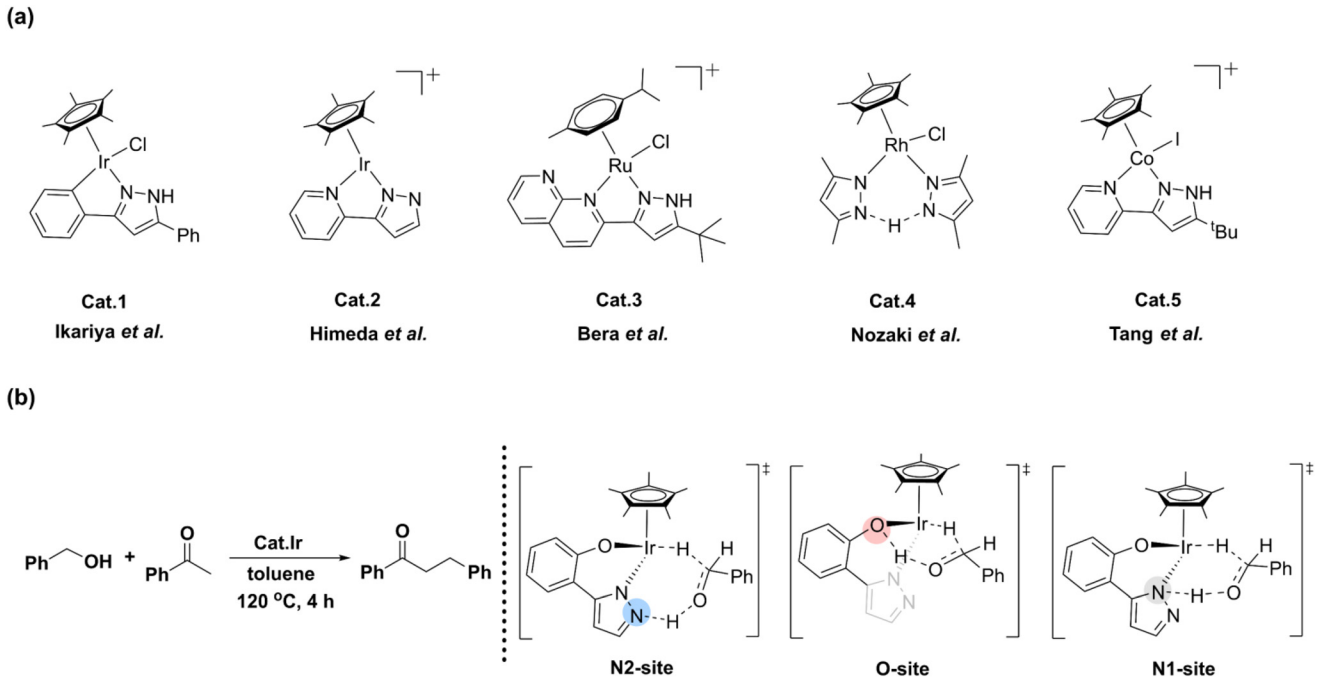

Fig. 1 (a) Metal–ligand cooperative catalysts with pyrazole ligands. (b) The borrowing hydrogen reaction catalyzed by an Ir MLC catalyst with triple ligand-assisted sites.

ligands (Fig. 1a, **Cat. 2**).[17] Under alkaline conditions, deprotonation of the OH and NH groups in the ligands not only enhanced electron donation to the metal center but also introduced proximal basic sites, thereby significantly improving catalytic activity in CO_2 hydrogenation. Studies revealed that heterolytic cleavage of H_2 is the rate-determining step, with the nitrogen site on the pyrazole ring substantially lowering the activation free energy, exemplifying typical metal–ligand cooperativity. In 2018, Bera *et al.* reported a Ru-based cooperative catalytic system supported by pyrazole ligands (Fig. 1a, **Cat. 3**), which efficiently catalyzed the dehydrogenation of primary amines to nitriles under mild, oxidant-free conditions.[18] Kinetic isotope effect studies indicated that cleavage of the N–H bond plays a key role in the rate of hydrogen release. Density functional theory (DFT) calculations revealed a hydride and proton transfer mechanism assisted by the β-N site of the pyrazole ligand, highlighting the crucial role of the ligand in the catalytic cycle. In 2021, Nozaki *et al.* synthesized a Cp*Rh complex supported by pyrazole ligands (Fig. 1a, **Cat. 4**), which efficiently catalyzed the dehydrogenation of dimethylamine-borane at room temperature.[19] The pyrazolate anionic ligand increased electron density at the metal center through its negative charge, while its nitrogen atom served as a proton-responsive Lewis basic site, cooperatively participating with Rh(III) in hydrogen transfer, thus enabling efficient catalysis. In 2024, Tang *et al.* reported a cobalt-centered *N*,*N*-bidentate complex (Fig. 1a, **Cat. 5**), capable of efficiently converting primary amines to nitriles under oxidant- and hydrogen acceptor-free conditions.[20] DFT calculations indicated that the β-N site on the pyrazole ligand cooperatively participates in proton transfer. The reaction mechanism includes both outer-sphere and inner-sphere dehydrogenation steps, with hydride transfer from the imine C–H bond to the cobalt center identified as the rate-determining step, fully demonstrating metal–ligand cooperativity. Although Ir, Rh, and Co complexes have dominated borrowing hydrogen studies, Earth-abundant transition metals such as Mn, Fe, and Ni have also been explored;[21–25] however, these studies remain limited in number and are largely at an early experimental stage. In summary, pyrazole ligands exhibit unique cooperative effects and regulatory potential in homogeneous catalysis by modulating the metal's electronic structure and facilitating proton transfer.

In homogeneous catalysis, the combination of experimental studies and theoretical calculations has long been a key approach for understanding reaction mechanisms and optimizing catalyst performance.[26–28] In recent years, predictive catalysis has gradually emerged as an important direction in this field,[29,30] with artificial intelligence techniques, particularly machine learning (ML), being recognized as a promising strategy for obtaining mechanistic insights and accelerating catalyst design. Poater *et al.* combined quantitative descriptors, such as %VBur, with DFT calculations and ML methods, which not only enabled the rationalization of known reaction mechanisms but also provided strong support for the rational design and screening of future catalysts.[31] Ke *et al.* employed machine learning to analyze hydrosilylation reactions, identifying the key factors influencing catalytic activity.[32] Currently, although applications in borrowing hydrogen reactions remain limited,[33–36] existing studies suggest that this strategy also holds significant potential. Machine learning can not only sys-

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tematically analyze large-scale catalyst datasets but also extract general structure–activity relationships, providing a theoretical basis for the development of efficient catalysts.[37]

Recently, Bagh *et al.* reported an air-stable iridium catalyst for the α-alkylation of ketones and arylacetonitriles with primary alcohols, demonstrating high efficiency and broad substrate scope.[38] This catalytic system operates effectively at low catalyst loading and enables high-yield conversion of diverse substrates, as well as the dehydrogenative coupling of 2-aminobenzyl alcohols with ketones to synthesize quinoline derivatives. Based on this system, the present work aims to systematically elucidate the mechanistic selectivity of the catalytic reaction, focusing on the key alcohol dehydrogenation step through density functional theory (DFT) calculations and investigating the roles of multiple potential cooperative sites on the ligand. To broaden the scope of the study, a comprehensive dataset comprising 126 catalytic systems was constructed, including Ir, Rh, and Co metal centers with diverse pyrazole ligand architectures. Machine learning (ML) methods were then applied to systematically analyze the critical geometric and electronic descriptors influencing activation energy barriers and catalytic efficiency, revealing intrinsic structure–performance relationships. By integrating theoretical mechanistic insights with data-driven analysis, this study not only deepens the understanding of metal–ligand cooperativity in borrowing hydrogen catalysis but also provides scientific guidance for the rational design of efficient and tunable catalysts.

Computational details

All DFT calculations in this study were carried out using Gaussian 16 (version A.03).[39] The geometries of all intermediates and transition states were optimized at the PBE0-D3 (BJ)[40,41]/def2-SVP[42] level of theory, followed by frequency calculations and intrinsic reaction coordinate (IRC) analyses to verify the nature of stationary points. To further improve the accuracy of energetic evaluations, solvent effects of toluene were modeled using the Solvation Model Density (SMD) implicit solvation model at the PBE0-D3(BJ)/def2-TZVP level.[43] A standard state free energy correction corresponding to the transfer of 1 mol of an ideal gas from 1 atm (32.26 L mol^{-1}) to 1 M (1 mol L^{-1}) concentration at 120 °C was applied to all species. Finally, to assess the reliability of our computational method, benchmark calculations were also performed using different functionals, including B3LYP-D3(BJ),[44–46] M06,[47] TPSSH[48] and WB97XD.[49] The Molecular Planarity Parameter (MPP)[50] analysis was performed using Multiwfn software (version 3.8).[51,52] Key intermediates were visualized and analyzed using VMD software.[53]

A comprehensive analysis and prediction of the dataset were conducted using various machine learning (ML) algorithms, including CatBoost,[54] Extreme Gradient Boosting (XGBoost),[55] Random Forest Regression (RFR),[56] Gradient Boost Regression (GBR),[57] Support Vector Regression (SVR),[58] Light Gradient Boosting Machine (LGBM),[59] Ridge Regression[60] and Linear Regression (LR).[61] All models were implemented with Python libraries such as Scikit-learn[62] to ensure reproducibility and transparency. Hyperparameter optimization was systematically performed *via* grid search coupled with five-fold cross-validation to enhance predictive performance and prevent overfitting. Model evaluation metrics included Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), with the formulas shown in (1), (2), and (3). To assess the robustness of the models, cross-validation was employed throughout the training process. Additionally, Principal Component Analysis (PCA)[63,64] was applied to reduce feature dimensionality and mitigate multicollinearity, retaining principal components that explain over 95% of the variance to preserve essential information. Meanwhile, Variance Inflation Factor (VIF)[65] analysis quantified feature correlations to optimize input variables. The impact of features on the output results is evaluated through the model's built-in feature importance and SHapley Additive exPlanations (SHAP) values.[66]

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{2}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3}$$

where y_i and $\hat{y}_i$ represent the calculated DFT and ML-predicted values, respectively; $\bar{y}$ denotes the average of the DFT dataset; and n signifies the sample size.

Results and discussion

To elucidate the mechanistic selectivity of the borrowing hydrogen reaction between alcohol and ketone catalyzed by Ir–pyrazole complexes, we systematically designed and investigated multiple plausible reaction pathways. Based on the proton-responsive sites within the ligand framework, three distinct pathways were considered, with the corresponding intermediates (IM) and transition states (TS) labeled using the prefixes a, b, and c, respectively. Specifically, pathway a involves the N2-site, representing the nitrogen atom located at the 2-position of the pyrazole ring; pathway b corresponds to the phenolic oxygen site (O-site) of the ligand; and pathway c involves the N1-site, representing the nitrogen atom at the 1-position of the pyrazole ring. The catalytic cycle of the Ir–pyrazole complex in this borrowing hydrogen reaction generally comprises three key steps: (1) dehydrogenation of benzyl alcohol to benzaldehyde, (2) aldol condensation between benzaldehyde and acetophenone to afford chalcone, and (3) hydrogenation of chalcone.

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Borrowing hydrogen reaction assisted by the N2 site

Based on pathway a assisted by the N2-site of the pyrazole ligand, the catalytic cycle commences from intermediate **IM0**. Initially, the Ir metal center in **IM0** coordinates with the substrate benzyl alcohol to form intermediate **IM1**, which is an endergonic process (ΔG = 5.5 kcal mol^{-1}), as shown in Fig. 2. The dehydrogenation of benzyl alcohol was investigated considering both the inner-sphere stepwise mechanism and outer-sphere cooperative hydrogen transfer mechanism. In the inner-sphere mechanism, the proton from benzyl alcohol is transferred to the pyrazole N2-site *via* a four-membered ring transition state **a-TS1** ($\Delta G^{\ddagger}$ = 11.2 kcal mol^{-1}), generating intermediate **a-IM2**. Subsequently, the hydride in **a-IM2** transfers to the Ir metal center through β-H elimination *via* transition state **a-TS2** ($\Delta G^{\ddagger}$ = 39.5 kcal mol^{-1}), forming the metal hydride species **a-IM3** and releasing benzaldehyde. The high activation free energy barrier of this step renders it kinetically unfavorable under typical conditions. Further consideration was given to the hydride transfer from the Ir center to the ligand O-site *via* transition state **a-TS6** ($\Delta G^{\ddagger}$ = 36.8 kcal mol^{-1}), leading to intermediate **a-IM4**; however, this pathway is similarly hindered by a high energy barrier, limiting its feasibility. In the outer-sphere mechanism, a concerted transfer of the proton and hydride from benzyl alcohol occurs *via* transition state **a-TS3** ($\Delta G^{\ddagger}$ = 18.3 kcal mol^{-1}), affording the metal hydride **a-IM3** and benzaldehyde. Given the significantly lower energy barrier, the dehydrogenation predominantly proceeds *via* this outer-sphere cooperative hydrogen transfer. Subsequently, benzaldehyde undergoes aldol condensation with acetophenone to form chalcone and water, with a reaction free energy of 2.0 kcal mol^{-1}. The hydrogenation of chalcone was then explored *via* two possible pathways involving hydride transfer to either the α-C or β-C of the C═C bond. For the β-C pathway, intermediate **a-IM3** proceeds through transition state **a-TS4** ($\Delta G^{\ddagger}$ = 25.2 kcal mol^{-1}), where the metal hydride and the proton on the pyrazole N2-site cooperatively transfer to the β-C and α-C, respectively, resulting in cleavage of the C═C bond. In contrast, the α-C pathway proceeds *via* transition state **a-TS5** ($\Delta G^{\ddagger}$ = 40.5 kcal mol^{-1}) involving a similar cooperative hydrogenation, but with a substantially higher activation free energy barrier. Therefore, chalcone hydrogenation preferentially occurs through the lower-energy transition state **a-TS4**, producing 1,3-diphenylacetone, while regenerating the active species **IM0** and completing the catalytic cycle. Overall, for pathway a assisted by the pyrazole N2-site, the rate-determining step in the borrowing hydrogen reaction is the outer-sphere cooperative hydrogenation *via* transition state **a-TS4**, with an energy barrier of 25.2 kcal mol^{-1}.

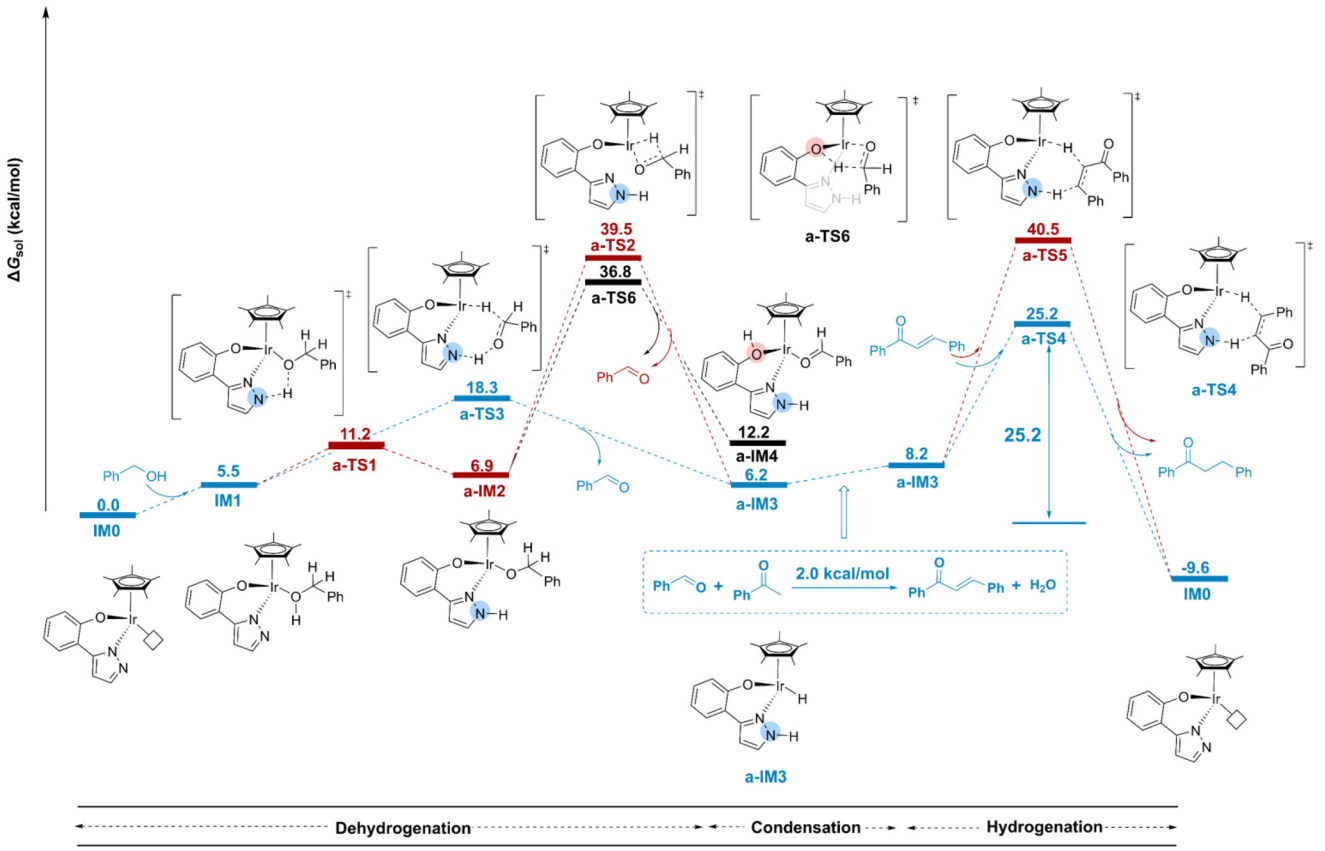

Fig. 2 The free energy diagram for the borrowing hydrogen reaction between alcohol and ketone catalyzed by an Ir–pyrazole complex with N2-site assistance.

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Borrowing hydrogen reaction assisted by the O-site

For the O-site assisted reaction pathway b, the catalytic cycle also initiates from intermediate **IM0**, as shown in Fig. 3. During the dehydrogenation of benzyl alcohol, no transition state corresponding to an inner-sphere proton transfer mechanism (**b-TS1**) was located. In contrast, the β-H elimination pathway involving hydride transfer *via* transition state **b-TS2** ($\Delta G^{\ddagger}$ = 32.5 kcal mol^{-1}) exhibits a relatively high activation free energy. Comparatively, the outer-sphere concerted hydrogen transfer mechanism *via* transition state **b-TS3** is more favorable, with a lower activation free energy of 25.4 kcal mol^{-1}, indicating a distinct kinetic advantage. Similar to reaction pathway a, two possible pathways were investigated for the hydrogenation of chalcone, involving hydride transfer from the metal center to either the α-C or β-C. Transition state **b-TS5**, corresponding to hydride transfer to the α-C, shows a prohibitively high activation free energy of 40.6 kcal mol^{-1}, making it kinetically unfavorable. In contrast, the concerted transfer of both the hydride and proton to the β-C and α-C, respectively, *via* transition state **b-TS4**, proceeds with a lower activation free energy of 32.6 kcal mol^{-1} and is thus more accessible. Therefore, for pathway b assisted by the O-site, the rate-determining step is the chalcone cooperative hydrogenation *via* transition state **b-TS4**, with an activation free energy barrier of 32.6 kcal mol^{-1}. This energy barrier is higher than the corresponding step in pathway a (**a-TS4**, 25.2 kcal mol^{-1}); thus pathway b is not favored.

Borrowing hydrogen reaction assisted by the N1-site

For pathway c assisted by the N1-site of the pyrazole ligand, the catalytic cycle similarly begins from intermediate **IM0**, as depicted in Fig. 4. The dehydrogenation of benzyl alcohol was investigated considering both the inner-sphere and outer-sphere mechanisms, analogous to pathways a and b. In the inner-sphere mechanism, intermediate **IM1** undergoes proton transfer to the pyrazole N1-site *via* transition state **c-TS1** ($\Delta G^{\ddagger}$ = 22.8 kcal mol^{-1}), forming intermediate **c-IM2**. Subsequently, β-H elimination transfers the hydride to the Ir metal center through transition state **c-TS2** ($\Delta G^{\ddagger}$ = 29.1 kcal mol^{-1}), yielding the metal hydride species **c-IM3** and releasing benzaldehyde. In the outer-sphere mechanism, intermediate **IM1** transfers both the proton and hydride simultaneously to the pyrazole N1-site and Ir metal center *via* transition state **c-TS3**, with an activation free energy barrier of 34.5 kcal mol^{-1}. Unlike pathways a and b, the higher barrier of the outer-sphere mechanism renders this route less favorable, and thus the dehydrogenation predominantly proceeds *via* the inner-sphere pathway. The hydrogenation of chalcone occurs *via* transition state **c-TS4** ($\Delta G^{\ddagger}$ = 40.6 kcal mol^{-1}), regenerating **IM0** and com-

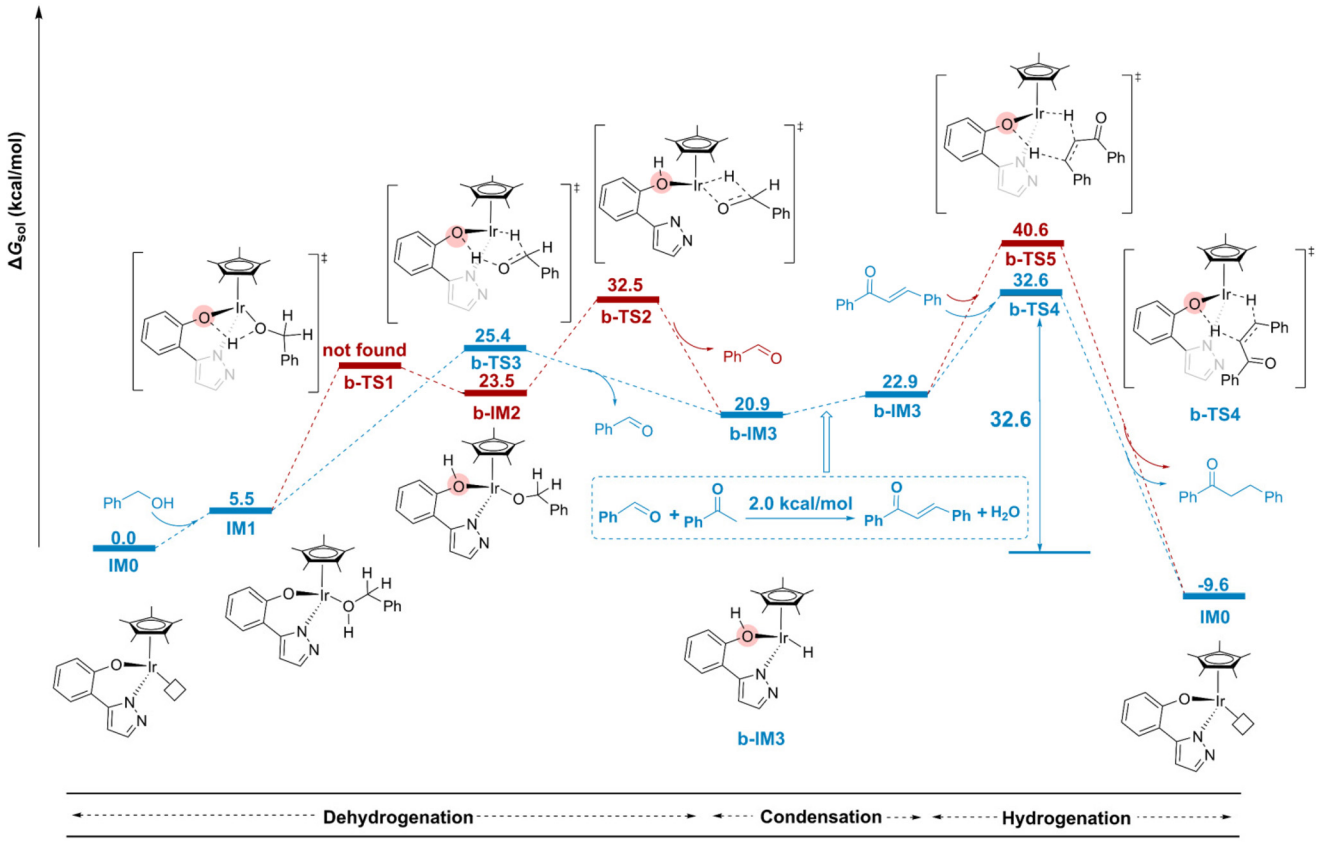

Fig. 3 The free energy diagram for the borrowing hydrogen reaction between alcohol and ketone catalyzed by an Ir–pyrazole complex with O-site assistance.

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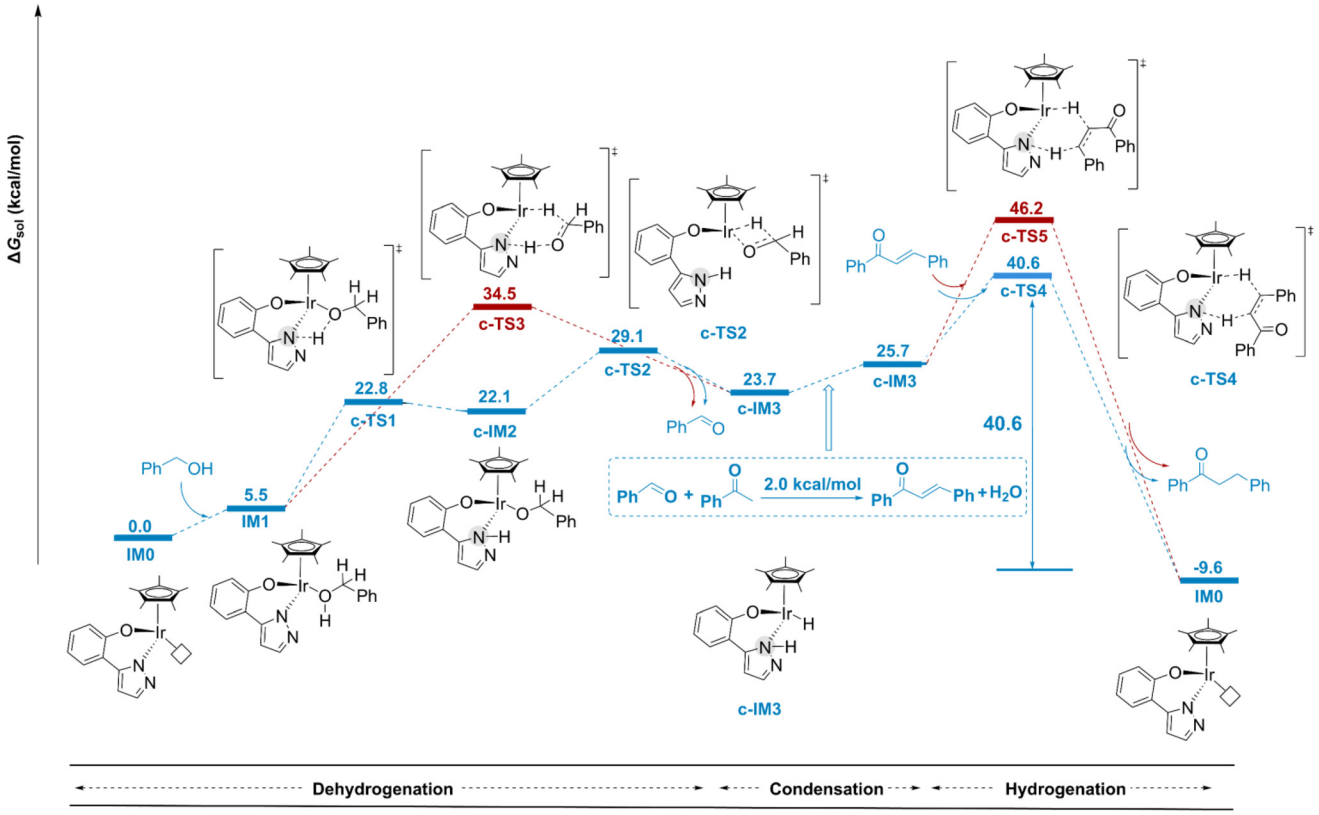

Fig. 4 The free energy diagram for the borrowing hydrogen reaction between alcohol and ketone catalyzed by an Ir–pyrazole complex with N1-site assistance.

pleting the catalytic cycle. For pathway c assisted by the pyrazole N1 site, the rate-determining step is also the chalcone hydrogenation *via* **c-TS4**, with an activation barrier of 40.6 kcal mol^{-1}. Its activation barrier is excessively high, rendering the reaction unlikely to proceed under practical experimental conditions. A comparative analysis of the rate-determining barriers for the three pathways (**a-TS4**: 25.2 kcal mol^{-1}, **b-TS4**: 32.6 kcal mol^{-1}, **c-TS4**: 40.6 kcal mol^{-1}) indicates that pathway a, assisted by the pyrazole N2-site, is the most favorable. Therefore, the borrowing hydrogen reaction catalyzed by the Ir–pyrazole complex preferentially follows pathway a.

The origin of the mechanistic preference

In the borrowing hydrogen reaction, an excessively endothermic dehydrogenation step can hinder the subsequent hydrogenation process; therefore, it is necessary to identify the key factors influencing this step. To elucidate the mechanistic selectivity, a distortion/interaction analysis[67] was conducted on the outer-sphere hydrogen transfer transition states **a-TS3**, **b-TS3**, and **c-TS3**, as illustrated in Fig. 5a. The results show that although the interaction energy of **a-TS3** (ΔE_{int} = −85.9 kcal mol^{-1}) is slightly less favorable than that of **b-TS3** and **c-TS3** (both ΔE_{int} = −103.5 kcal mol^{-1}), the substrate distortion energy associated with **a-TS3** is significantly lower (**a-TS3**: $\Delta E_{dist\text{-}sub}$ = 69.0 kcal mol^{-1}; **b-TS3**: $\Delta E_{dist\text{-}sub}$ = 94.8 kcal mol^{-1}; **c-TS3**: $\Delta E_{dist\text{-}sub}$ = 98.4 kcal mol^{-1}). Consequently, **a-TS3** exhibits the lowest activation electronic energy. In contrast, **c-TS3** shows the highest activation barrier due to the substantial distortion energies of both the substrate and the catalyst. These findings indicate that the assistance of the pyrazole N2-site in pathway a effectively reduces the substrate distortion, thereby lowering the overall reaction barrier and favoring the dehydrogenation process. To elucidate the mechanistic selectivity of the inner-sphere and outer-sphere pathways in the N2-assisted route, we performed a molecular planarity parameter (MPP) analysis on **a-TS2** and **a-TS3**, as shown in Fig. 5b. The atom of the structure is coloured according to the span of deviation from the plane (SDP) value and the shade of the colour represents the degree of deviation from the fitting plane. Atoms above and below the fitted plane are shown in red and blue, respectively. The results show that **a-TS2** has the highest MPP value (0.536 Å), indicating significant deviation from planarity, whereas **a-TS3** exhibits a smaller MPP value (0.187 Å), consistent with the |φC1–C2–C3–N1| dihedral angle analysis. The more distorted ligand structure in **a-TS2** leads to a higher energy, resulting in a higher activation free energy compared to **a-TS3** and thus favoring the outer-sphere pathway.

Machine learning model evaluation

To gain deeper insight into the factors influencing the activity of this class of catalysts, this study integrates DFT calculations with ML methods to construct structure–performance relation-

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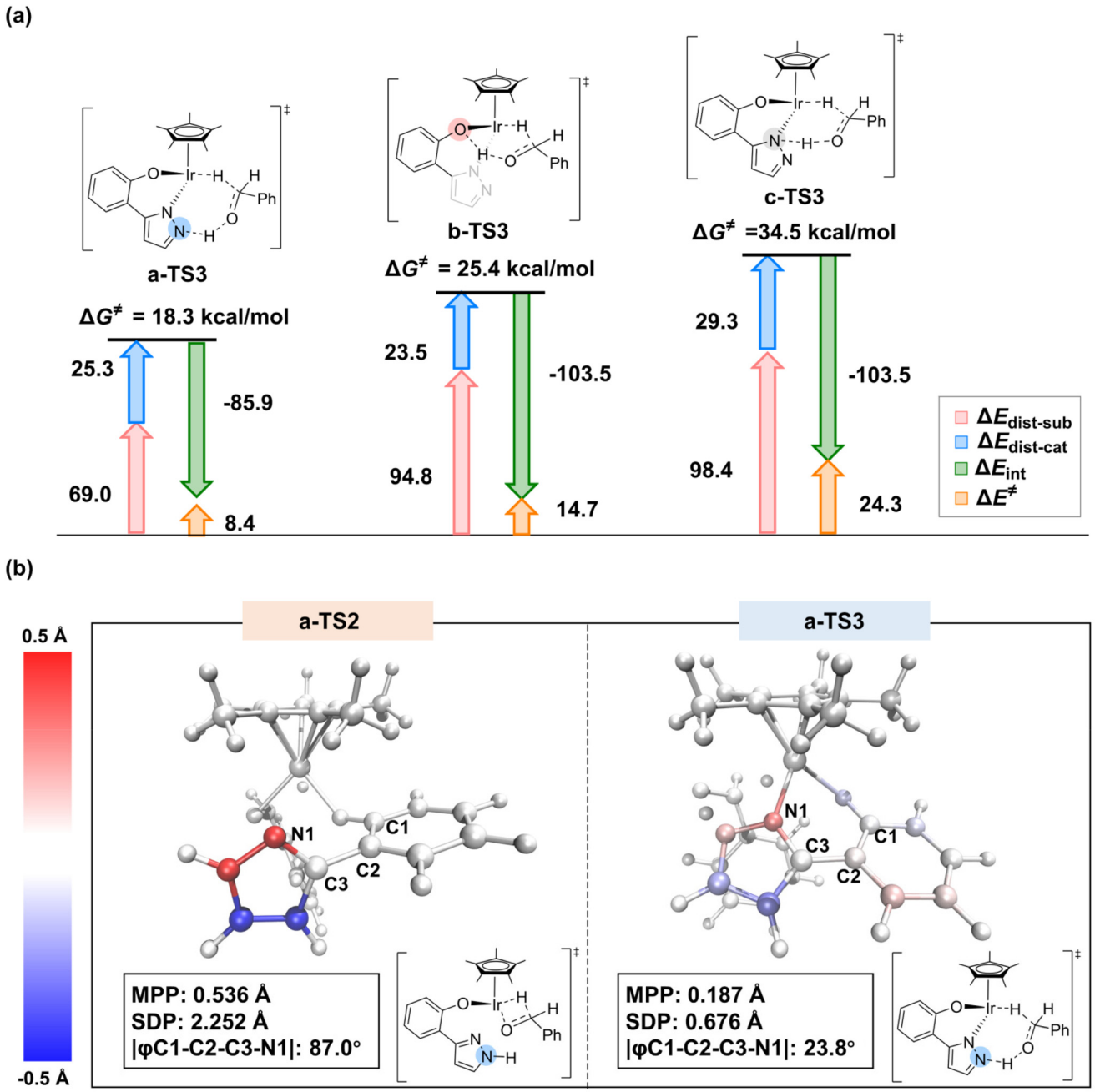

Fig. 5 (a) Distortion/interaction analysis. (b) Molecular planarity parameter analysis.

ship models. The ML workflow, shown in Fig. 6d, sequentially includes data collection and preprocessing, model and feature selection, determination of the optimal model, and feature importance analysis. Focusing on the cooperative hydrogen transfer process (Fig. 6a), we obtained the reaction free energy barriers for the step from intermediates to transition states *via* DFT calculations. Based on these data, a dataset comprising 126 catalytic systems was constructed, with metal centers and ligand structures designed as illustrated in Fig. 6b. The dataset was derived from representative catalytic systems reported in the literature.[68–77] Although it does not cover all possible ligand variations, it exhibits good diversity in terms of metal types, ligand structures, electronic properties, and steric effects, thereby effectively capturing the key features that influence catalytic activity. Feature selection is critical to ML model performance; selected features should be readily accessible and effectively characterize both the geometric and electronic properties of the catalysts. Accordingly, a series of descriptors were extracted, mainly categorized into two types as shown in Fig. 6c: (1) electronic descriptors, including the metal center charge (q_M), the charge at the reactive site N2 (q_{N2}), the highest occupied molecular orbital energy (E_{HOMO}), and the HOMO–

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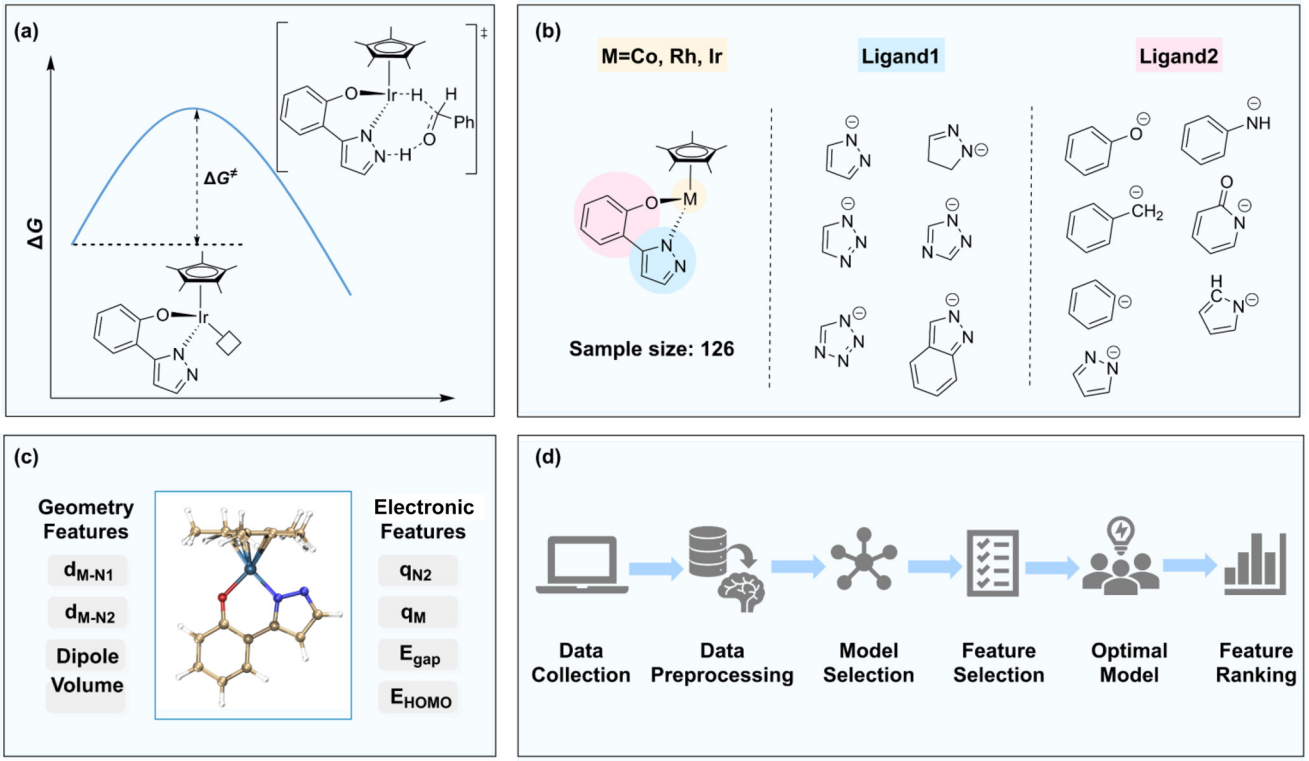

Fig. 6 (a) The free energy barrier of the outer-sphere hydrogen transfer process. (b) Sample space. (c) Different descriptors of the intermediate complexes. (d) Machine learning process.

LUMO gap (E_{Gap}), reflecting electronic distribution and transition capability; and (2) geometric descriptors, including the distance between the metal and the N1 atom (d_{M-N1}), the distance between the metal and the N2 atom (d_{M-N2}), the dipole moment (Dipole) and the molecular volume (Volume), which characterize the spatial configuration of the catalyst. These descriptors were rationally designed based on relevant literature and chemical intuition,[78–80] and their validity will subsequently be verified through feature importance analysis in machine learning models, ensuring that they can reliably and effectively capture the structure–activity relationships of the catalysts.

To eliminate redundant correlations among features, pairwise relationships were evaluated using the Pearson correlation coefficient (PCC),[81] as shown in Fig. 7a. A PCC value of ±1 indicates a perfect positive or negative correlation, with significant correlations highlighted in red and blue. The results show that most feature pairs exhibit |PCC| values below 0.8, suggesting the absence of severe multicollinearity. Therefore, these features are suitable as independent input variables for model construction. The dataset was split into 80% for training and 20% for testing, a commonly adopted ratio that balances stability and practical applicability. Accordingly, this partitioning strategy was employed throughout model training and validation in this study.

In this study, we systematically evaluated the performance of eight commonly used regression models and combined this with feature importance analysis to identify key factors influencing the catalyst activation free energy. The models assessed include CatBoost, XGBoost, RFR, GBR, SVR, LGBM, Ridge Regression, and LR. Model performance was evaluated using R^2, RMSE, and MAE metrics, with detailed parameters and results provided in Tables S3 and S4 in the SI, where R^2 indicates the proportion of variance explained by the model, with values closer to 1 representing better fit. The results indicate that XGBoost and CatBoost significantly outperform the other models. XGBoost exhibits an extremely low training error (Train RMSE = 0.0712, Train MAE = 0.0556, Train R^2 = 0.9999), while the test error is noticeably higher (Test RMSE = 1.5829, Test MAE = 1.2268, Test R^2 = 0.9572), indicating that the model exhibits some degree of overfitting on the training set and reflecting its limited generalization ability. In contrast, CatBoost demonstrated superior generalization ability, as shown in Fig. 7b. CatBoost yielded a test R^2 of 0.9570, a test RMSE of 1.5858, and a test MAE of 1.2687, with a small gap between training error (Train RMSE = 0.5818) and test error, indicating a more robust model with lower overfitting risk. Fig. 7c compares the R^2 performance across models; while GBR and RFR performed well on the training set, their test set performance was inferior, indicating clear overfitting or underfitting. Traditional regression methods such as LR and SVR show certain stability with small sample sizes but lack sufficient fitting capability for this task, resulting in larger prediction errors and difficulty capturing the complex nonlinear

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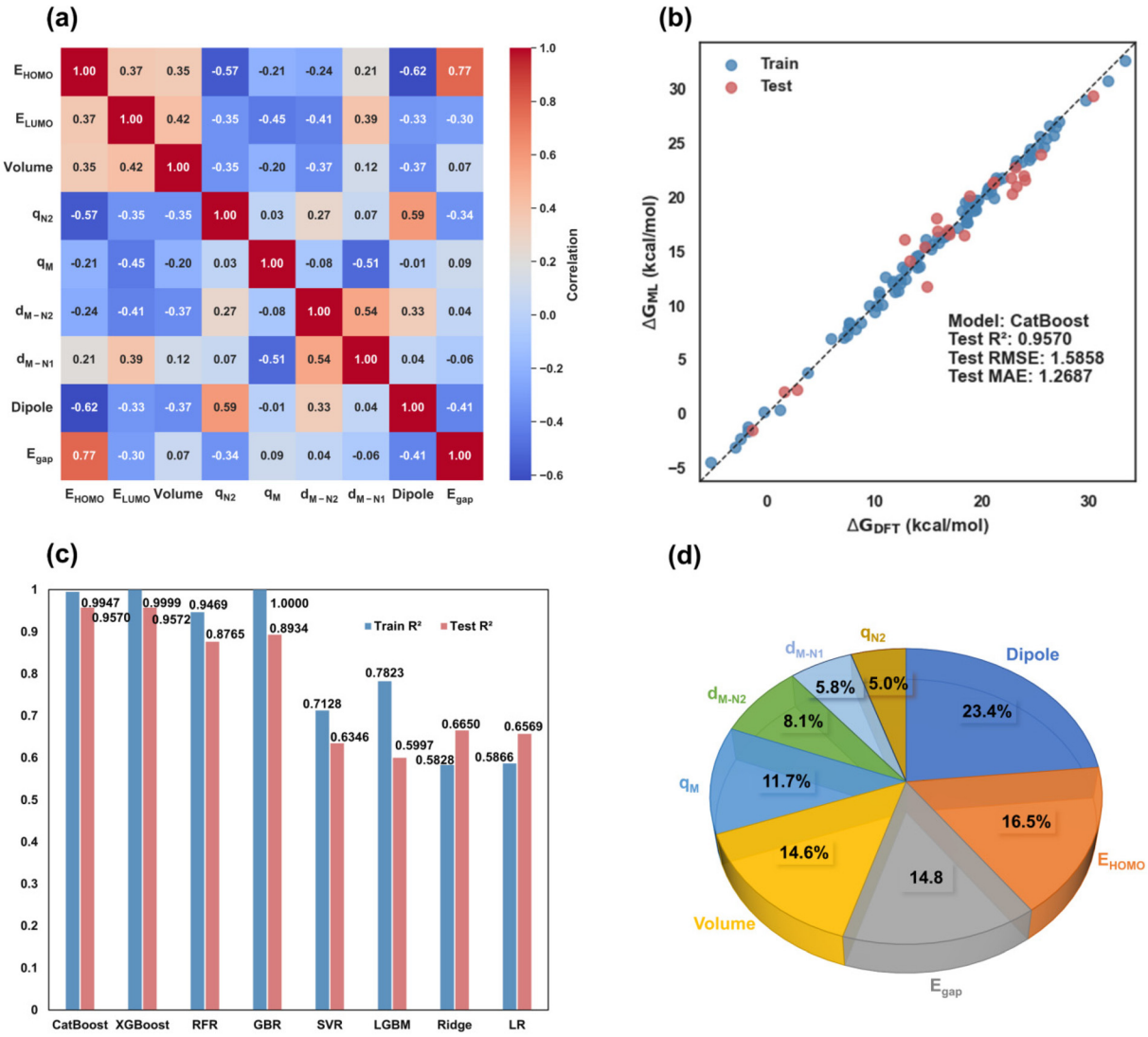

Fig. 7 (a) Feature correlation heatmap. (b) Performance of the optimal model. (c) R^2 comparison across models. (d) SHAP feature importance pie chart.

relationships between descriptors and activation free energy. In contrast, tree-based ensemble learning models such as XGBoost and CatBoost possess stronger feature learning and generalization capabilities. Notably, CatBoost further mitigates overfitting through categorical feature handling and gradient direction regularization, achieving an optimal balance between stability and performance, and was therefore selected as the preferred model in this study.

To elucidate the influence of various features on catalyst performance, this study employed the SHapley Additive exPlanations (SHAP) method to generate a feature importance pie chart, quantitatively illustrating the relative contribution of each input feature to the model predictions. As shown in Fig. 7d, each sector of the pie chart corresponds to a feature, with the sector size proportional to its cumulative SHAP value, providing an intuitive representation of its importance in the model. The analysis revealed that the dipole moment (Dipole) and the highest occupied molecular orbital energy (E_{HOMO}) are the two most critical features affecting the activation free energy, indicating that electronic distribution and frontier orbital characteristics play dominant roles in regulating catalytic activity. A larger dipole moment enhances the catalyst's polarity, which facilitates stabilization of polar transition states and thus promotes the activation and transformation of polar substrates such as alcohols at the catalytically active site. Given that the transition state corresponds to a dehydrogenation reaction wherein the ligand site donates electrons to accept protons, a higher HOMO energy generally corresponds to a stronger electron-donating capability, thereby facilitating the hydrogen transfer process. Additionally, molecular volume also significantly influences the prediction outcomes, suggesting that the spatial configuration of the catalyst plays an important role in reaction pathway selection and transition state stabilization. These feature contributions are highly con-

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sistent with the DFT mechanistic analysis, wherein pathway a exhibits lower distortion energy and stronger metal–ligand orbital cooperation, resulting in superior catalytic performance. In summary, the SHAP analysis not only clarifies the key structural factors governing catalytic activity but also provides valuable theoretical guidance for the rational design and efficient screening of future catalysts.

Conclusion

This study integrates DFT calculations with ML techniques to systematically elucidate the key reaction pathways and mechanistic selectivity of alcohol borrowing hydrogen (BH) reactions catalyzed by Ir(III)–pyrazole complexes. Three possible pathways were proposed: N2-site-assisted, O-site-assisted, and N1-site-assisted. DFT calculations revealed that the BH reaction preferentially proceeds *via* the N2-site-assisted pathway, with an activation free energy of 25.2 kcal mol^{-1}. Distortion/interaction analysis of the dehydrogenation step indicated that distortion energy plays a decisive role. Furthermore, an ML model was constructed to accurately predict the effects of different metals and ligands on the BH reaction barrier. A CatBoost model incorporating eight descriptors characterizing the electronic and structural features of the catalysts achieved excellent predictive accuracy (R^2 = 0.9570) and strong generalization performance. Feature importance analysis identified the dipole moment, HOMO energy level, and molecular volume as the main factors influencing the reaction barrier, highlighting the critical roles of frontier orbital characteristics, electronic distribution, and spatial configuration in modulating catalytic reactivity. Overall, this work not only provides theoretical insight into the mechanism of Ir(III)–pyrazole-catalyzed BH reactions but also, through the application of ML modeling, reveals key structural–electronic factors governing catalytic activity, thereby offering valuable guidance for the rational design of efficient and tunable metal–ligand cooperative catalysts.

Author contributions

Hui-Qi Mo: investigation, data curation, writing – original draft. Cheng Hou: conceptualization, writing – review & editing, supervision, funding acquisition.

Conflicts of interest

The authors declared that they have no conflicts of interest to this work.

Data availability

Additional data are made available in the SI tables of this manuscript.

Supplementary Information includes: (1) comparison of free energy barriers using different functionals; (2) calculation details of standard state correction; (3) substrate scope investigation; (4) DFT-calculated energies of all optimized structures; (5) optimized hyperparameters for each machine learning model; (6) performance of ML models on train and test sets; (7) comparison of ML predictions with DFT-calculated ΔG; (8) machine learning sample dataset; (9) code and scripts for model training and performance evaluation; and (10) DFT-calculated energies of all optimized structures in the sample dataset. See DOI: **https://doi.org/10.1039/d5qo01139f**.

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